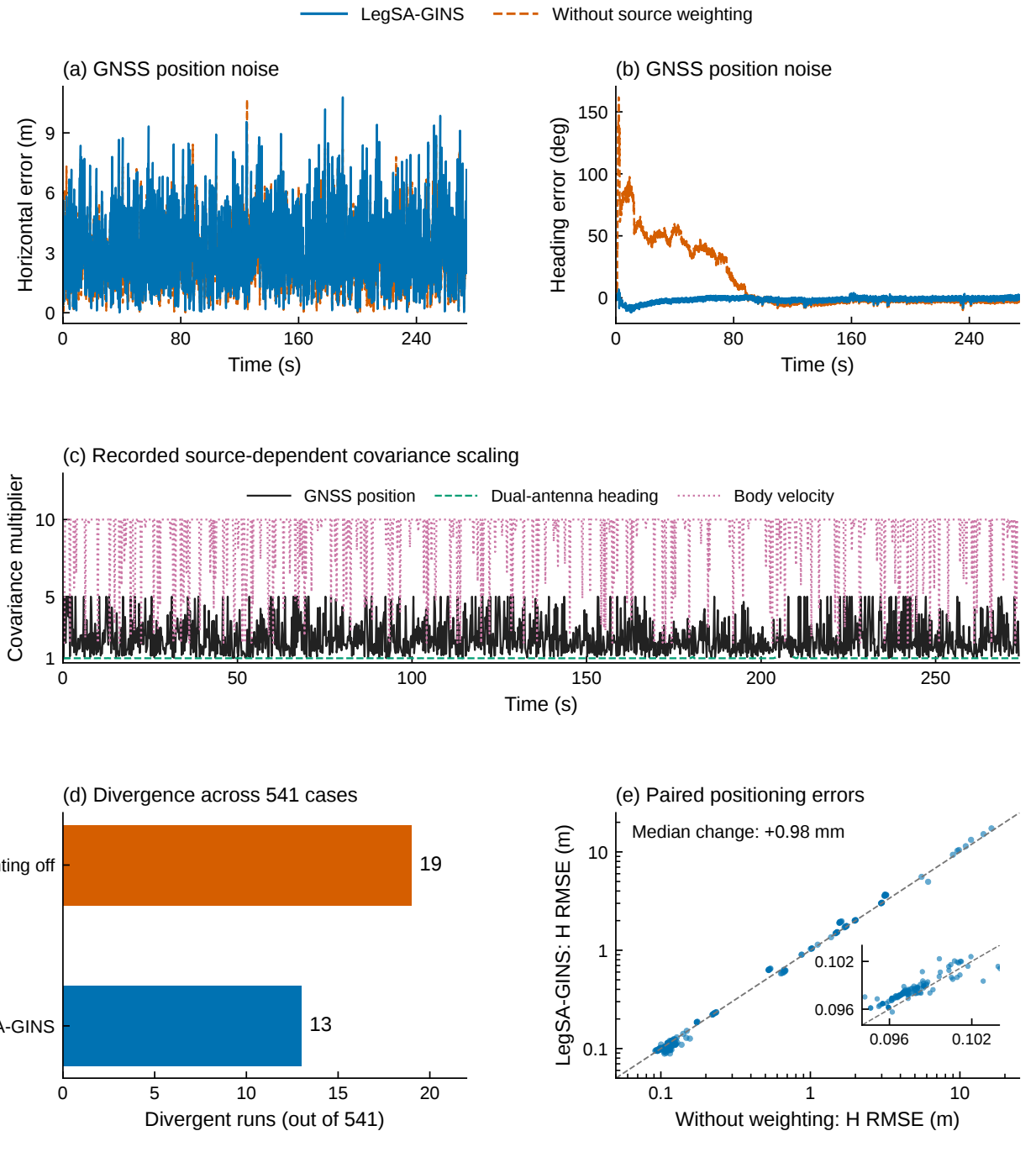

LegSA-GINS
Without source weighting
(a) GNSS position noise
Horizontal error (m)
0
3
6
9
0
80
160
240
Time (s)
(b) GNSS position noise
Heading error (deg)
0
50
100
150
0
80
160
240
Time (s)
(c) Recorded source-dependent covariance scaling
GNSS position
Dual-antenna heading
Body velocity
Covariance multiplier
1
5
10
0
50
100
150
200
250
Time (s)
(d) Divergence across 541 cases
Weighting off
LegSA-GINS
19
13
0
5
10
15
20
Divergent runs (out of 541)
(e) Paired positioning errors
Median change: +0.98 mm
LegSA-GINS: H RMSE (m)
0.1
1
10
0.096
0.102
0.1
1
10
Without weighting: H RMSE (m)